

Base-Paper Candidates for a CSE Numerical-Analysis-and-Simulation Course Project

TL;DR

- The strongest, most feasible base papers cluster around four syllabus topics that map cleanly onto the student's AI/ML-CV-systems strengths: **spline interpolation inside Kolmogorov-Arnold Networks (KANs)**, **Runge-Kutta/ODE solvers inside diffusion samplers and Neural ODEs**, **Levenberg-Marquardt/Gauss-Newton least squares inside SLAM/bundle adjustment**, and **Newton-Raphson inside power-flow analysis** — each uses a syllabus numerical method as an isolatable subroutine an undergraduate team can reproduce at small scale.
- The single best "scaffold" candidate is **KAN-ODEs (Koenig, Kim & Deng, CMAME 2024)** because it stacks TWO syllabus topics (B-spline interpolation + RK ODE integration) in one reproducible pipeline — its 240-parameter Lotka-Volterra demo runs on a single CPU core — and the choice of B-spline order/grid and RK solver is under-justified and easily swapped.
- Prioritize the cross-domain-application direction (the student's preferred #1): take the paper's numerical subroutine (e.g., the bisection MPP locator, the LM bundle-adjustment step, the RK diffusion sampler) and re-apply it to a different dataset/domain; use the methodological-extension direction (swap in an alternative syllabus method and compare convergence/accuracy/cost) as the fallback.

Key Findings

- Across ~18 web searches plus a targeted sub-investigation, the numerical methods with the richest recent engineering literature that fits the "subcomponent, not headline" framing are: **spline interpolation (KANs)**, **Runge-Kutta / ODE integration (diffusion + Neural ODE)**, **nonlinear least squares via Gauss-Newton/Levenberg-Marquardt (SLAM/vision)**, **Newton-Raphson (power flow)**, **Monte Carlo (structural reliability)**, **Metropolis-Hastings (Bayesian NN uncertainty)**, **finite difference (PINNs)**, **eigenvalue methods (spectral GNNs)**, and **direct linear solvers LU/Gauss-Jordan (circuit simulation)**.
- The best undergraduate-feasible scaffolds are those where the numerical step runs on a tiny problem (Lotka-Volterra, a 5-bus grid, a 48-image test set, an MNA matrix of a small RLC circuit) even though the original paper also ran large experiments.
- Many papers make an explicitly "standard/off-the-shelf" numerical choice — e.g., DROID-SLAM "uses the Gauss-Newton algorithm for simplicity" (verbatim from Bundle Adjustment in the Eager Mode), diffusion works defaulting to a "second-order midpoint solver," KANs defaulting to 3rd-order B-splines — which are exactly the under-justified choices the syllabus wants students to interrogate by testing alternatives.
- Bias toward the student's strengths is well-served: KANs, diffusion samplers, Neural ODEs, SLAM/bundle adjustment, 3D Gaussian Splatting, spectral GNNs, LLM speculative decoding, and Bayesian NN uncertainty are all AI/ML/CV/systems-native.

Details

Group 1 — Spline interpolation / curve fitting (student's core AI/ML strength)

1. KAN: Kolmogorov–Arnold Networks (base architecture paper)

- Citation: Liu, Wang, Vaidya, Ruehle, Halverson, Soljačić, Hou, Tegmark. "KAN: Kolmogorov–Arnold Networks." arXiv:2404.19756, 2024 (ICLR-linked). <https://arxiv.org/abs/2404.19756>
- Contribution: Replaces MLP fixed activations with learnable univariate functions on edges, parameterized as B-splines, giving competitive accuracy with fewer parameters and better interpretability.
- Numerical method: **Spline interpolation** — each learnable activation is a B-spline, $\text{spline}(x) = \sum c_i B_i(x)$, with a spline order (default cubic, $k=3$) and grid of knots. This is the syllabus "spline interpolation" topic sitting at the very core of the model.
- Scaffold assessment: Very strong and isolatable. The spline machinery can be studied on a single KAN layer fitting a 1-D or 2-D function on a laptop; no large dataset needed. The choice of cubic B-splines and uniform grid is a standard/off-the-shelf default, explicitly a good target for testing alternatives (the literature already spawned FastKAN with Gaussian RBFs, rKAN rational functions, Chebyshev-KAN, SineKAN).
- Suggested angle: **Methodological extension** — at the KAN activation step, replace cubic B-spline interpolation with alternative syllabus interpolants (Lagrange polynomial, Newton's divided-difference polynomial, natural cubic spline with different knot placement) and compare fitting RMSE, training stability, and cost on a small regression benchmark. Alternatively **cross-domain**: apply a small KAN to a Bangladeshi dataset (e.g., load/traffic/weather time series) and analyze how spline grid refinement affects extrapolation.

2. Kolmogorov–Arnold Networks (KANs) for Time Series Analysis

- Citation: Vaca-Rubio, Blanco, Pereira, Caus. "Kolmogorov–Arnold Networks (KANs) for Time Series Analysis." 2024 IEEE Globecom Workshops; arXiv:2405.08790. <https://arxiv.org/abs/2405.08790>
- Contribution: Applies KANs to satellite-traffic forecasting, outperforming MLPs with fewer parameters. ([Semantic Scholar](#)) ([arxiv](#))
- Numerical method: **Spline interpolation** (spline-parametrized univariate activations, with order k and G intervals controlling the interpolation between control points). ([arXiv](#))
- Scaffold assessment: Strong. Small forecasting dataset, clear ablation of KAN spline parameters already in the paper — reproducible at small scale. The spline order/grid is an off-the-shelf default.
- Suggested angle: **Cross-domain application** — port the KAN forecaster to a different engineering time series (grid load, network traffic, air-quality sensors) and run a controlled study of spline grid size vs forecast error; a natural undergraduate-scale contribution.

Group 2 — ODEs: Runge-Kutta, Euler, finite difference (student's CV/generative strength)

3. KAN-ODEs (TWO syllabus topics — top overall pick)

- Citation: Koenig, Kim, Deng. "KAN-ODEs: Kolmogorov-Arnold Network Ordinary Differential Equations for Learning Dynamical Systems and Hidden Physics." *Computer Methods in Applied Mechanics and Engineering* (CMAME), Vol. 432, 117397, 2024; arXiv:2407.04192. <https://arxiv.org/abs/2407.04192>
- Contribution: Uses a KAN as the gradient-getter inside a Neural ODE to learn dynamical systems and hidden physics with fewer parameters and better interpretability than MLP Neural ODEs. ([ResearchGate](#))
- Numerical method: **Both spline interpolation (KAN B-splines) AND numerical ODE integration (Runge-Kutta / adaptive ODE solver)** — the KAN outputs du/dt and a classical ODE solver integrates it forward in time. ([ScienceDirect](#))
- Scaffold assessment: Excellent. The flagship demo is the **Lotka-Volterra predator-prey model**, which is tiny and runs on a laptop: on a single CPU core the 240-parameter KAN-ODE reached 2.6×10^{-5} training loss in just 10^4 epochs, already beating the best 10^5 -epoch MLP Neural ODE result of 3.0×10^{-5} (an effective 3–4 $\times$ speedup at $\sim$ equal parameter count of 240 vs 252). Burgers' and Fisher-KPP are optional stretch goals. The ODE solver choice is a standard adaptive solver default — a clean swap target. (Funded under NSF CBET-2143625.)
- Suggested angle: **Methodological extension** — at the ODE-integration step, compare Euler vs RK4 vs adaptive RK and measure accuracy/stability/cost of the learned Lotka-Volterra dynamics; at the KAN step, swap spline basis. **Cross-domain**: fit KAN-ODEs to a different small dynamical system (SIR epidemic curve, RC circuit, pendulum) and study solver sensitivity.

4. STORK: Faster Diffusion and Flow-Matching Sampling (RK ODE solvers in diffusion)

- Citation: Tan, Wang, Bertozzi, Ryu. "STORK: Faster Diffusion and Flow Matching Sampling by Resolving both Stiffness and Structure-Dependence" (Stabilized Taylor Orthogonal Runge-Kutta). arXiv:2505.24210v2, Oct 2025. <https://arxiv.org/abs/2505.24210> (code: github.com/ZT220501/STORK)
- Contribution: A training-free, structure-independent diffusion/flow-matching ODE solver built on a class of stiff-ODE stabilized Runge-Kutta methods, targeting the mid-NFE regime (20–50 NFE) used by models such as Stable Diffusion 3.5, FLUX, and SANA.
- Numerical method: **Runge-Kutta ODE solvers** for the probability-flow ODE; explicitly frames the problem in terms of classical numerical-analysis stiffness (Burden & Faires cited).
- Scaffold assessment: Moderate–strong. The full diffusion model is heavy, but the **sampler is training-free and swappable** — a team can take a small pretrained diffusion model (e.g., CIFAR-scale or a toy 2-D distribution) and study samplers without training. Prior work (DPM-Solver, Karras EDM) provides small reference implementations.

- Suggested angle: **Methodological extension** — at the sampling step, compare Euler vs Heun (RK2) vs RK4 vs the paper's stabilized RK on a small/2-D diffusion model, measuring sample quality (FID or a 2-D distance) vs number of function evaluations. This directly instantiates the syllabus RK/step-size/stiffness content.

5. Neural ODEs / RK4 for time-series forecasting

- Citation: e.g., "Fractional Neural Ordinary Differential Equations for Time-Series Forecasting," *Electronics* 15(9):1929, 2025 (adopts a fixed-step RK4 solver); and the broad Neural-ODE literature (Chen et al. 2018) using Euler/RK4/Dormand-Prince.
<https://doi.org/10.3390/electronics15091929>
- Contribution: Continuous-depth models parameterizing dz/dt with a neural network, integrated by an ODE solver.
- Numerical method: **Euler / Runge-Kutta (RK4) / Dormand-Prince ODE integration** as the forward pass ("a fixed-step fourth-order Runge-Kutta (RK4) solver is adopted").
- Scaffold assessment: Strong; small forecasting datasets, and the solver is an explicit, isolatable choice.
- Suggested angle: **Methodological extension** comparing solver order/step-size on forecasting accuracy and cost; **cross-domain** to a Bangladeshi sensor/engineering series.

Group 3 — Nonlinear least squares / optimization: Gauss-Newton & Levenberg-Marquardt (student's CV strength)

6. Bundle Adjustment in the Eager Mode

- Citation: "Bundle Adjustment in the Eager Mode." arXiv:2409.12190, 2024.
<https://arxiv.org/abs/2409.12190>
- Contribution: A PyTorch-native, differentiable bundle-adjustment library using nonlinear least-squares solvers integrated into eager-mode deep learning.
- Numerical method: **Levenberg-Marquardt / Gauss-Newton nonlinear least squares** (solving normal equations $(J^T W J) \delta x = -J^T W \Delta z$), combining the syllabus's linear-system, optimization, and curve-fitting content.
- Scaffold assessment: Strong and highly isolatable. The LM/GN solver can be built and tested on a tiny synthetic 2-view or 3-point reconstruction. The paper explicitly flags an under-justified competitor choice (verbatim): "DROID-SLAM ... uses the Gauss-Newton algorithm for simplicity and implements CUDA kernels from scratch specific to its use case. However, the solution is not generalizable to all settings nor easy to re-build." — an ideal invitation to compare alternatives.
- Suggested angle: **Methodological extension** — at the BA optimization step, compare plain Gauss-Newton vs Levenberg-Marquardt (with varying damping λ) vs gradient descent on a small synthetic reprojection problem: convergence rate, robustness to poor initialization, final reprojection error. **Cross-domain**: apply the same LM least-squares fitting to a non-vision curve-fitting problem (sensor calibration).

7. Camera Pose Revisited (PnP with least-squares optimization)

- Citation: "Camera Pose Revisited" (PnP-ProCay78). arXiv:2601.12567 / Applied Sciences 16(6):2690. <https://arxiv.org/abs/2601.12567>
- Contribution: A planar Perspective-n-Point algorithm combining Cayley rotation parameterization with nonlinear least-squares minimization of projection residuals, plus a deterministic initialization scheme.
- Numerical method: **Least-squares optimization** (nonlinear least-squares of reprojection residuals) + the syllabus "curve fitting/least squares" and "root of optimality conditions" content.
- Scaffold assessment: Strong. The datasets are ordinary RGB/thermal calibration images; the numerical core is small and reproducible.
- Suggested angle: **Methodological extension** — study how initialization and the least-squares solver choice affect convergence; **cross-domain** to a different calibration rig.

Group 4 — Root finding: Newton-Raphson, bisection, false position (electrical/power + CV)

8. Newton-Raphson power flow (data-driven initialization)

- Citation: Yan, Vazinram, Kaseb, et al. "Data driven approach towards more efficient Newton-Raphson power flow calculation for distribution grids." arXiv:2504.11650, 2025. <https://arxiv.org/abs/2504.11650>
- Contribution: Improves Newton-Raphson power-flow convergence by predicting good initial guesses (analytical voltage bounds, supervised/PINN models, RL).
- Numerical method: **Newton-Raphson root finding** (the core PF solver) plus Jacobian linear solves.
- Scaffold assessment: Strong. Benchmark grids (IEEE test systems) are public and small; NR itself is textbook. The initialization is the novelty, so NR is a clean, isolatable subroutine.
- Suggested angle: **Methodological extension** — compare Newton-Raphson vs Gauss-Seidel vs fast-decoupled load flow on a small IEEE bus system, studying iteration count/convergence vs initialization. **Cross-domain**: apply NR root-finding to a different nonlinear engineering system.

9. Power Flow Analysis of a 5-Bus Power System Based on Newton-Raphson

- Citation: Nwachukwu. "Power Flow Analysis of a 5-Bus Power System Based on Newton-Raphson Method." arXiv:2506.23425, 2025. <https://arxiv.org/abs/2506.23425>
- Contribution: A didactic 5-bus power-flow study comparing NR (quadratic convergence) with a MATLAB implementation validated against PowerWorld.
- Numerical method: **Newton-Raphson** (and comparison to Gauss-Seidel).
- Scaffold assessment: Excellent for undergraduates — explicitly small (5-bus), reproducible, with reference tooling. Almost a ready-made lab.

- Suggested angle: **Methodological extension** — extend the comparison to bisection/secant on a scalar sub-problem, or false-position vs NR on the mismatch equations; or **cross-domain** to a DC microgrid.

10. Fast Otsu Thresholding Using Bisection Method (from targeted sub-investigation)

- Citation: Kodathala. "Fast OTSU Thresholding Using Bisection Method." arXiv:2509.16179, 2025 (cs.CV). <https://arxiv.org/abs/2509.16179>
- Contribution: Accelerates Otsu image thresholding by replacing the exhaustive 256-level search with a bisection search over the (empirically unimodal) between-class variance. Experimental validation on 48 standard test images shows a 91.63% reduction in variance computations and 97.21% reduction in algorithmic iterations vs exhaustive search — reducing 256 evaluations to an average of 21.4 per image — with exact threshold matches in 66.67% of cases and 95.83% within 5 gray levels. (arXiv + 2)
- Numerical method: **Bisection method** (adapted from root-finding to maximum-finding); explicitly cites Numerical Recipes and Burden & Faires.
- Scaffold assessment: Excellent and the cleanest to reproduce — 48 public grayscale test images, a few dozen lines of Python. Note the caveat that bisection-on-a-unimodal-maximum is closer to ternary/golden-section search than to true sign-change root-finding.
- Suggested angle: **Methodological extension** — compare bisection vs golden-section search vs exhaustive Otsu for accuracy/speed on a small image set (directly instantiates syllabus root-finding + golden-section optimization). **Cross-domain**: apply the same bracketing search to another unimodal threshold-selection problem (edge detection, audio VAD).

11. Improved MPPT using numerical analysis techniques for PV systems (from targeted sub-investigation)

- Citation: Guanghai, Siddiqui, Aman, et al. "Improved maximum power point tracking algorithms by using numerical analysis techniques for photovoltaic systems." Results in Engineering (Elsevier), Vol. 21, 101740, 2024. <https://doi.org/10.1016/j.rineng.2024.101740>
- Contribution: MPPT algorithms based on numerical-analysis techniques (predictor-corrector, compared against bisection and perturb-and-observe) that converge faster and oscillation-free to the PV maximum power point.
- Numerical method: **Bisection and false-position (regula falsi) root finding** used as the MPP locator on the PV power-voltage curve.
- Scaffold assessment: Strong. The MPP-finding step runs on the standard single-diode PV I-V model with textbook parameters — fully separable from any hardware rig, implementable in MATLAB/Python/Simulink.
- Suggested angle: **Methodological extension** — compare bisection vs false-position vs Newton-Raphson vs golden-section for locating the MPP on the single-diode curve: convergence speed, oscillation, robustness. Very clean numerical-methods project. **Cross-domain**: reuse the bracketing on a different single-peak optimization.

Group 5 — Direct linear solvers: LU decomposition, Gauss / Gauss-Jordan elimination (from targeted sub-investigation)

12. Fast Circuit Simulation of Linear Electronic Circuits using Modified Nodal Analysis

- Citation: Voyer, Nakabayashi, Kondo. "Fast Circuit Simulation of Linear Electronic Circuits using Modified Nodal Analysis." 2024 IEEE ECCE, pp. 3825–3832.
DOI:10.1109/ECCE55643.2024.10860273. (Extended journal version in IEEE Trans. Power Electronics, Feb 2026.) <https://doi.org/10.1109/ecce55643.2024.10860273>
- Contribution: Closed-form (sum-of-weighted-exponentials) waveform expressions for linear circuits described in Modified Nodal Analysis (MNA) form. ([ResearchGate](#))
- Numerical method: **Gauss-Jordan elimination and LU decomposition** applied to the MNA matrix ("two Gauss-Jordan elimination steps"; section "Lower-Upper (LU) decomposition"). ([ResearchGate](#))
- Scaffold assessment: Very undergraduate-friendly. A small RLC circuit's MNA matrix can be built by hand (Kirchhoff laws) and solved with Gauss-Jordan/LU — classic circuit-analysis coursework. Off-the-shelf elimination is the mechanical engine, not the novelty.
- Suggested angle: **Methodological extension** — compare Gauss elimination vs Gauss-Jordan vs LU decomposition (with/without partial pivoting) on MNA matrices of small circuits: operation count, numerical stability, round-off error. **Cross-domain:** apply the same solvers to a small finite-element or structural stiffness system.

13. An Efficient Transient Nonlinear Circuit Simulator (LU inside a preconditioner)

- Citation: Zhang, Lin, Chen, Cai (Guangdong University of Technology). "An Efficient Transient Nonlinear Circuit Simulator Using Exponential Integration and Block-Jacobi Precondition." arXiv:2509.16219, 2025. <https://arxiv.org/abs/2509.16219>
- Contribution: A transient nonlinear circuit simulator combining a 2nd-order exponential integrator with block-Jacobi-preconditioned GMRES; reports a $1.95\times$ – $3.27\times$ speedup in total computation time vs Backward Euler with Inexact Newton, with time steps reduced by an average of 60.70% (up to 74.59%). ([arxiv](#))
- Numerical method: **LU decomposition** to invert diagonal sub-blocks in the preconditioner ("we adopt the strategy of LU factorization ... stable with high computational efficiency"); also uses Newton-Raphson. ([arxiv](#))
- Scaffold assessment: The LU-on-small-dense-block step is fully separable and reproducible; the large benchmark circuits are not needed to study it. **Caveat:** the v2 (Nov 20, 2025) of this preprint was withdrawn — treat with care and prefer paper #12 for a stable direct-solver base.
- Suggested angle: **Methodological extension** — study LU factorization reuse and pivoting on small block matrices; compare direct LU vs iterative solve as block size grows (illustrates why the paper switches to GMRES at scale).

Group 6 — Monte Carlo, Metropolis-Hastings, and other feasible extras

14. Reliable uncertainty estimates in deep learning with efficient Metropolis-Hastings algorithms

- Citation: Nature Communications, s41467-026-70015-z (integrates Metropolis-Hastings acceptance steps into stochastic-gradient HMC for Bayesian NNs).
<https://www.nature.com/articles/s41467-026-70015-z>
- Numerical method: **Metropolis-Hastings / MCMC** (syllabus Monte Carlo + Metropolis-Hastings).
- Scaffold assessment: The MH accept/reject step is isolatable and can be studied on a small BNN or toy posterior. A good fit for the student's ML strength.
- Suggested angle: **Methodological extension** — study how the MH proposal distribution and acceptance rate affect posterior calibration on a small regression BNN; **cross-domain** to uncertainty quantification on a small engineering regression dataset.

15. Monte Carlo structural reliability (civil)

- Citation: "Probabilistic Modeling and Structural Reliability based Monte Carlo Simulation: A Case Study," IJETT 72(5):321-331, 2024 (<https://ijettjournal.org/archive/ijett-v72i5p133>); and "An Improved Monte Carlo Reliability Analysis Method Based on BP Neural Network," MDPI Applied Sciences 15(8):4438, 2025 (<https://www.mdpi.com/2076-3417/15/8/4438>).
- Numerical method: **Monte Carlo simulation + random number generation** for probability-of-failure estimation.
- Scaffold assessment: Excellent for undergraduates — small limit-state functions, clear convergence plots; reproducible with a few thousand samples (the IJETT case runs 5,000 then 10^6 trials).
- Suggested angle: **Methodological extension** — compare crude Monte Carlo vs variance-reduction (importance sampling, Latin hypercube) on a reinforced-concrete beam limit state, studying convergence of the failure probability; **cross-domain** to a different limit-state (pressure vessel, slope stability).

Honorable mentions (feasible, student-aligned)

- **Physics-Informed Neural Networks with finite-difference PDE loss** — "An Improved Physics-Informed Neural Network Approach for Solving the FitzHugh-Nagumo Equation," MDPI Computation 13(12):275, 2025 (compares PINN vs explicit finite difference on a 1-D PDE); and Lim/Dutta/Rotaru, "Physics Informed Neural Network using Finite Difference Method," IEEE SMC 2022 — syllabus **finite difference method**, very reproducible.
- **3D Gaussian Splatting** (Kerbl et al., "On the Error Analysis of 3D Gaussian Splatting," arXiv:2402.00752) — syllabus **gradient optimization + error analysis**; the SGD optimization of Gaussian parameters and the projection error analysis are isolatable, though full training is compute-heavy (use a small scene).

- **Improving Expressive Power of Spectral GNNs with Eigenvalue Correction** (AAAI 2024, arXiv:2401.15603) — syllabus **eigenvalue decomposition (power method/QR)** on the graph Laplacian; the eigen-computation is isolatable on small graphs.
- **A Parallel-in-Time Newton's Method for Nonlinear MPC** (arXiv:2409.20027) — syllabus **Newton's method for optimization / linear solves**; small nonlinear control examples are reproducible.

Recommendations

1. **First choice — KAN-ODEs (paper #3).** It is the richest single scaffold because it hits two syllabus topics at once (B-spline interpolation + RK ODE integration), its flagship 240-parameter Lotka-Volterra demo runs on a single CPU core, and both numerical choices (spline order/grid, ODE solver) are under-justified defaults ideal for the methodological-extension direction. Concrete plan: reproduce the Lotka-Volterra KAN-ODE; then (a) swap the ODE solver among Euler/RK2/RK4/adaptive and tabulate accuracy-vs-cost, and (b) swap the KAN spline basis order/grid and measure fit quality. Benchmark that would change the recommendation: if reproducing KAN-ODE training proves unstable on your hardware within ~2 weeks, fall back to #9 or #10.
2. **Safest/most reproducible choice — 5-Bus Newton-Raphson (paper #9) or Fast Otsu Bisection (paper #10).** Both are explicitly small, have public data/tooling, and give a clean root-finding comparison (NR vs Gauss-Seidel vs bisection/secant; bisection vs golden-section vs exhaustive). Choose #10 if the team wants a CV flavor, #9 for an electrical flavor.
3. **Best CV-strength stretch — Bundle Adjustment in the Eager Mode (paper #6).** Build a tiny synthetic bundle-adjustment problem and compare Gauss-Newton vs Levenberg-Marquardt (varying damping) vs gradient descent — directly interrogating DROID-SLAM's "Gauss-Newton for simplicity" choice.
4. **Best generative-AI stretch — STORK diffusion sampler (paper #4).** Use a small/2-D or CIFAR-scale pretrained diffusion model and compare Euler/Heun/RK4/stabilized-RK samplers on quality-vs-NFE. Only pursue if a pretrained model is available (avoid training from scratch).
5. **Staged execution:** Week 1-2, reproduce the paper's numerical subroutine on the smallest example. Week 3-4, implement 2-3 alternative syllabus methods at that step. Week 5-6, run the convergence/accuracy/cost/stability comparison and (if going cross-domain) port to a second dataset. Decision threshold: if you cannot isolate and re-run the numerical subroutine standalone by end of Week 2, switch to a more isolatable paper (#9/#10/#12/#15).
6. **Prefer cross-domain application (the stated #1 preference) when the subroutine is trivially portable** (bisection MPP locator → other unimodal search; Monte Carlo limit state → new structure; LM least squares → sensor calibration); use methodological extension when the subroutine is tightly coupled to the pipeline (KAN splines, diffusion sampler).

Caveats

- Several candidates are **arXiv preprints** (KAN, STORK, Bundle Adjustment in Eager Mode, the transient circuit simulator, Fast Otsu, the two power-flow papers) and may not be peer-reviewed. If the course requires a peer-reviewed venue, prefer KAN-ODEs (CMAME), the Vaca-Rubio KAN paper (IEEE Globecom Workshops), the MPPT paper (Results in Engineering), the MNA circuit paper (IEEE ECCE), the Metropolis-Hastings paper (Nature Communications), the spectral-GNN paper (AAAI), and the Monte Carlo reliability papers (IJETT/MDPI).
- **Paper #13's preprint v2 (Nov 20, 2025) was withdrawn** — use #12 as the primary direct-solver base paper and treat #13's numbers cautiously.
- A few "improvement" papers report **speedups and speculative benefits** (e.g., STORK fidelity gains, the transient-simulator $1.95\text{--}3.27\times$ claim, speculative-decoding throughput); treat headline performance numbers as authors' claims, not independently verified, and focus the project on the numerical subroutine rather than reproducing the headline result.
- The Fast Otsu paper's use of "bisection" is, strictly, closer to ternary/golden-section search on a unimodal function than to sign-change root-finding — worth stating explicitly in your report (and itself a good discussion topic).
- Some source quotes for the MNA circuit paper (#12) came from a ResearchGate preview with imperfect OCR of math symbols; the method names (Gauss-Jordan, LU) are unambiguous but verify exact equations against the IEEE version.
- Full reproduction of large diffusion, 3D Gaussian Splatting, or LLM speculative-decoding systems is **not** feasible on undergraduate compute; only pursue those if you restrict to the isolatable numerical subroutine on a small/toy instance.
- Deliberately deprioritized: pure benchmarking/comparative-study papers (e.g., "Comparative Study on Reliability Estimation Using Monte Carlo Simulation," arXiv:2201.08013), per the task's instruction that generic benchmarking is the lowest-priority direction.